Project report submitted for
Banana Spoilage Detection
Machine Learning
(UML501)

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Abstract

Approximately one-third of the food produced globally is wasted throughout the supply chain, translating into significant financial and resource losses. To mitigate this issue, the present work develops a state-of-the-art multimodal deep learning system that enables non-destructive and highly reliable assessment of banana ripeness and early spoilage. Bananas are the fourth most important food crop worldwide and are highly perishable, making them a perfect candidate for exploring automated quality monitoring solutions. The architecture synthesizes four synergistic data modalities: (i) RGB/multispectral imaging interpreted via specialized CNN models to extract ripeness-related visual features, (ii) hyperspectral reflectance profiles spanning 410–940 nm to quantify internal biochemical transitions, (iii) VOC sensor outputs to detect spoilage precursors, and (iv) ambient variables,including temperature, humidity, and pressure that modulate the ripening trajectory. By using a confidence-weighted late fusion technique, our framework combines the outputs from image, spectral, and VOC models, while merging environmental features directly with sensor data. The system categorizes bananas into four distinct groups -unripe, ripe, overripe, and rotten, which are evaluated through metrics like accuracy, precision, recall, and F1-score. The fusion of multiple sensing modalities enables the architecture to retain stable performance despite challenges such as sensor noise, gradual calibration drift, and partial data loss. The multimodal fusion mechanism contributes to a consistent performance gain of approximately 2% when compared to traditional models that operate on a single modality. In addition, the framework has been carefully engineered for scalability, interpretability, and seamless real-world deployment. Overall, the integration of these components establishes the system as a significant milestone in the field of intelligent food quality monitoring, while simultaneously promoting the adoption of sustainable and technology-centric agricultural solutions.

1. Introduction

1.1 Background

Food security and sustainability have become urgent global issues in recent years. Reports from the Food and Agriculture Organization (FAO) show that nearly one-third of all food produced worldwide, around 1.3 billion tonnes each year, is either lost or wasted throughout the supply chain. A significant part of this loss happens due to post-harvest decay in fruits and vegetables. This not only results in economic losses worth hundreds of billions of dollars, but also leads to harmful environmental effects, like unnecessary greenhouse gas emissions, excessive water use, and land degradation.

Fresh fruits are especially prone to high post-harvest losses because of their fragile nature, short shelf life, and strict market standards that often reject even slightly imperfect produce. Traditional quality assessment methods mainly rely on manual inspection and destructive testing. Trained personnel examine fruit batches based on visual traits such as color, texture, and firmness, using subjective judgment and handling. Although expert assessment has its benefits, traditional inspection methods have clear drawbacks, like variable results between operators, reduced reliability due to fatigue, and high costs that limit scalability in industrial environments.

The ripening of climacteric fruits involves a complex mix of biochemical processes that affect various quality attributes. For instance, the breakdown of chlorophyll leads to visible color changes, the conversion of starch to sugars affects sweetness, changes in pectin concentration alter texture and firmness, and the release of volatile compounds impacts flavor and overall consumer perception. Unfortunately, traditional inspection methods cannot assess these internal biochemical changes without physically damaging the fruit, nor can they give a complete and objective measure of overall quality.

1.2 Motivation

Among the many important fruits grown around the world, bananas (Musa spp.) are particularly suitable for research on automated freshness assessment. They are the fourth most valuable food crop globally, after wheat, rice, and maize, and make up about 16% of total world fruit production [7]. Nearly 90% of banana cultivation is found in Asia, Latin America, and Africa. This crop is not just a daily nutritional staple for millions; it also serves as an important economic resource for farming communities in developing regions. Nutritionally, bananas provide significant benefits. They contain potassium, vitamin B6, vitamin C, dietary fiber, and natural sugars, making them a key part of balanced diets for diverse populations.

Despite their value, bananas are highly perishable and prone to rapid quality deterioration during post-harvest handling, transport, and storage. Being climacteric fruits, they continue to ripen after harvest due to autocatalytic ethylene production, which drives a succession of physiological and biochemical transformations [8]. The ripening process is typically divided into seven identifiable stages, each marked by specific peel color changes (from green to yellow and eventually brown-black), variations in firmness, sweetness, and aroma. Under ambient environmental conditions, the fruit may progress from peak ripeness to a commercially unacceptable state within just a few days unless storage parameters are tightly maintained.

Economic and Technological Rationale

The financial impact of banana spoilage is substantial. Both under-ripe bananas, which lack the characteristic sweetness and flavor, and over-ripe bananas, which display brown spotting, excessive softening, or initial decay, lose market value sharply. Consumer purchasing behavior is further influenced by appearance, with visually imperfect fruits often rejected despite acceptable internal quality [9]. Factors such as temperature instability, humidity fluctuations, physical bruising, and transport delays can accelerate degradation and propagate losses throughout the supply chain.

There is an urgent need in the industry for reliable, cost-effective, and non-destructive systems that can monitor banana ripeness continuously. Moreover, technologies that work well for bananas can be applied to other perishable fruits. This could lead to better post-harvest management practices in agriculture.

Technological Imperative: Need for Multi-Modal Approaches

Current automated quality assessment systems mainly use single-sensor methods, such as visual imaging, spectroscopy, or gas sensing. Each method has its strengths, but none is enough on its own. Visual imaging cannot detect internal biochemical changes until external symptoms appear. Spectroscopy needs expensive equipment and special handling. Gas-based measurements alone do not provide the complete classification needed for accurate ripeness prediction [10].

This research is based on the idea that combining multiple sensing techniques like visual imaging, hyperspectral reflectance, volatile compound detection, and monitoring environmental factors through smart sensor fusion algorithms can effectively address the limitations of single-modality systems. This approach aims to achieve higher accuracy, earlier spoilage predictions, and greater reliability in the varied conditions of real-world industrial environments.

1.3 Objective

The main goal of this research is to create, implement, and test a deep learning framework for classifying banana ripeness and detecting early spoilage. This framework will combine four different data streams using adaptive fusion methods.

Specific Objectives:

1. **Develop High-Performance Visual Classification Models**: Evaluate and optimize deep learning models for classifying banana ripeness using RGB images. We will start with baseline methods like logistic regression and move through custom CNNs to the latest transfer learning techniques, like EfficientNet, aiming for over 97% classification accuracy.
2. **Optimize Numerical Sensor Data Processing:** Evaluate machine learning algorithms such as SVM, Random Forest, ANN, KNN, Gradient Boosting, XGBoost, and 1D CNN with multimodal numerical features like hyperspectral data, VOC, and environmental measurements. Our goal is to identify the best classifiers for determining ripeness using sensor data.
3. **Design Confidence-Weighted Multimodal Fusion Architecture:** Implement late fusion methods that combine visual and numerical predictions based on real-time confidence levels. This will improve the system's ability to handle sensor noise, drift, and partial data loss.
4. **Achieve Early Spoilage Detection:** Show that the framework can detect spoilage signs before visible symptoms occur by using chemical and spectral markers that appear before changes in color and texture.
5. **Validate Performance Improvements Over Single-Modality Baselines:** Provide evidence that multimodal fusion achieves statistically significant accuracy gains of 2-4% compared to individual modality models through thorough comparative evaluation.
6. **Assess Practical Deployment Feasibility:** Examine the computational efficiency, scalability, interpretability, and real-world usability of the proposed framework for integration into agricultural supply chain operations.

This study aims to advance the domain of intelligent food quality monitoring through the following key contributions:

1. Developing an integrated multimodal assessment framework that brings together visual, spectral, chemical, and environmental sensing for reliable ripeness evaluation.
2. Presenting a structured methodology for step-by-step model development and performance refinement, enabling transparent tracking of system improvements.

3. Demonstrating adaptive sensor-fusion strategies that improve accuracy and operational stability under real, variable deployment conditions.
4. Establishing reproducible evaluation baselines by releasing and benchmarking on a large-scale banana ripeness dataset consisting of 13,478 images.
5. Delivering practical guidance for industrial adoption, particularly for cold-storage operations, supply-chain distribution points, and retail settings where freshness monitoring plays a crucial economic role.

2. Related Works

This section examines previous work in fruit ripeness detection, offering a summary according to both sensing modality and methodological approach, assessing strengths and weaknesses, and situating our multimodal framework in relation to existing work.

2.1 Visual Fruit Classification using Deep Learning

Due to their proficiency in automatically extracting hierarchical visual features, convolutional neural networks (CNNs) have become the dominant strategy for fruit ripeness classification. Studies using transfer learning with pretrained networks (i.e., ResNet, EfficientNet, GoogleNet) consistently report high accuracy, often above 96-99% on banana datasets, with careful data augmentation [15]. Raghavendra et al. employed CNN+MLP framework that achieved 98.4% accuracy in classifying banana ripeness, combining convolutional feature extraction with classification heads in a multilayer perceptron. Likewise, a transfer learning approach based on GoogleNet classified banana freshness stages with around 98.9% accuracy. These results illustrate how CNNs fine tuned to fruit datasets with aggressive data augmentation produce robust generalization allowing for reliable visual assessments of fruit ripeness stages often with fine distinctions between preferred ripe, overripe, and early decay.

Extensive systematic reviews of over 1,500 published studies showed that CNNs are more accurate than traditional machine learning techniques (Support Vector Machines, conventional Artificial Neural Networks) in analyzing large image datasets, whereas traditional machine learning techniques were somewhat better at analyzing lower dimensional data collected from sensors [15]. Additionally, visual only approaches have limitations: they are unable to capture internal biochemical changes before visual symptoms are exhibited, they are unreliable in variable lighting conditions, and they may misclassify fruit with cosmetic defects that do not affect internal quality.

2.2 Volatile Organic Compounds and Chemical Sensing

Metal-oxide volatile organic compound (VOC) sensors and electronic noses (e-nose) can provide advance warning of rotting fruit by detecting biochemical changes that occur before visual deterioration. An array of TGS and SGP sensors has been employed to monitor volatiles (e.g., ethanol) and biochemical changes associated with ripening in bananas [12].

Chen and colleagues constructed a dual MOS electronic nose/camera system to demonstrate that VOC profiles of fruit provide extremely informative signatures that can be employed to classify fruit ripeness.

In one study, a neural network that was trained on VOC sensor data achieved 100% classification accuracy classifying unripe, ripe, and rotten bananas. In another study, a DenseNet method had an accuracy of 97%, utilizing VOC "smell fingerprints" of mixed fruits.

Despite their strong performance, there are challenges associated with using gas sensing systems in practice. Sensor drift will generally require frequent recalibration over time, sensor readings are influenced by ambient environmental factors (such as temperature and humidity), and reading raw sensor outputs requires complex, time-consuming, and advanced pattern-recognition algorithms. These limitations suggest the need to incorporate complementary modalities, rather than using it alone.

2.3 Hyperspectral and Spectral Imaging

Spectral reflectance in the visible and near-infrared bands (410-940 nm) are specific yet detailed biochemical fingerprint that can reveal internal composition of fruit that cannot be seen with standard RGB cameras. For example, Rajkumar et al. (2016) showed that hyperspectral imaging can provide much higher discrimination between subtle ripeness stages of bananas by capturing differences in sugar accumulation, moisture distribution, and starch degradation (11). Hyperspectral systems use an optical sensor that allows the computational feature extraction performed with various convolutional neural network architectures, resulting in high accuracy for detecting fine physiological stages and spoiled fruit before visual symptoms start to occur. On the other hand, deploying hyperspectral systems involves factors such as significant capital investment ($3,000-$10,000+ per unit), requires training of the operator(s), involves complicated calibrating processes, and conducts complex preprocessing for noise (all of which are barriers for use in agricultural environments where resources are limited).

2.4 Integration of Wireless Sensor Networks and Internet of Things

Wireless sensor network (WSN) architectures emerged as workable approaches for continuous environmental monitoring in storage facilities. Altaf et al. made observations of Xbee-based WSN sensing implementations for efficient banana ripening monitoring efforts by disbursing a number of WSN sensing nodes that were able to monitor temperature, humidity, ethylene concentration, and CO_2 concentration in areas of storage [6, 13]. WSNs take advantage of statistical artificial neural networks to correlate environmental parameters with fruit physiological condition and subsequently convey actionable assessments related to status and predictions for actions.

Recent work looked at IoT-enabled solutions based on using ESP32 microcontrollers with multi-sensor arrays and demonstrated good ripeness prediction accuracy using ensemble algorithms such as CatBoost at a relatively low cost of implementation [14]. This recent work suggests that it is reasonably possible to design and deploy low-cost, inexpensive sensor networks for monitoring agricultural quality, however simple real-time environmental sensing by itself is limited to the information it can provide without visual or chemical analytical information.

2.5 Multimodal Fusion Approaches

The utilization of multi-modal data types, including spectral measurements, volatile organic compound (VOC) sensors, visual imaging, and environmental data shows consistent improvement over single data-type methods. In studies on both bananas and tomatoes, those systems show 2-5% absolute accuracy improvements over unimodal approaches while offering higher resilience to sensor failure and uncertainty in measurement noise.

The most effective implementations utilize late-fusion and adaptive weighting systems that allow different types of data to contribute based on their confidence level which are adjusted every time the sensory data is collected. These models efficiently capitalize on the synergistic effects of visual, chemical, and spectral observations representing the most current state-of-the-art for accurate measurements for early spoilage detection. However, there are persistently daunting issues including time synchronization across heterogeneous sensors, long-term sensor drift, improved gracefulness in performance when sensors fail or suffer other quality degradation, and overall computational efficiency in real time.

2.6 Addressing the Challenges of Deploying Models to Real-World Contexts

While laboratory results are impressive, getting these models to work in the wild presents another formidable challenge. Variance in lighting conditions, occlusions, sensor drift, and noise can all disrupt the function of trained models (Liu et al, 2022). To address this, Liu et al. presented low cost models for detection that used generative adversarial networks (GANs) and attention based techniques to build robustness within their models. More broadly, Olisah and colleagues created ensemble classifiers using convolutional neural networks (CNNs) and streams of different inputs to improve robustness and detection of subtle characteristics of ripeness in the context of changing environmental parameters.

2.7 Research Gap and Our Contribution

An examination of the literature reveals several barriers to the development of reliable automated ripeness assessment systems. First, most studies only utilize two or three sensing modalities (e.g., visible imagery, near infrared reflectance, volatile organic compounds, environmental factors, etc.), with most studies carrying out any reasonable analysis of more than two or three sensing modalities being largely absent. The opportunity for a more integrated assessment that incorporates not only visual information and hyperspectral signatures, but also the analysis of volatile organic compounds and environmental measurements would allow a more holistic examination of fruit physiology and the ripening process. Most of the systems reported in the literature are focused on classifying visible ripeness stages and have not been designed for or have not given attention to the early detection of microbial spoilage before evident external manifestations. Early prediction is essential for quality control located before storage and during transit, and before the detection of visible spoilage symptoms. A second point of concern is that most of the research groups using multi-sensor evidence fusion, glue the data together, average the data, or simply concatenate the data. Importantly, they are missing the opportunity to use the data on an as needed basis, in particular, considering the unpredictable fluctuations in individual sensor reliability. This missing opportunity is thanks to developing a weight dependent on their established confidence/credibility in a certain predictive context. A third potential lack is that most study prototypes reported in this area (fruit ripeness) do not consider practical issues (proper calibration, scalability, computational burden, and process explainability) for moving towards an application.

This research addresses the identified gaps through the following contributions:

1. Development of a comprehensive multimodal deep learning framework that simultaneously incorporates four complementary sensing modalities: visual imaging, hyperspectral data, volatile organic compound (VOC) measurements, and environmental parameters.
2. Implementation of a confidence-weighted late fusion strategy that adaptively adjusts the contribution of each modality based on real-time reliability, improving robustness under variable operating conditions.
3. Focus on early spoilage prediction, leveraging chemical and spectral markers to detect internal quality degradation before visible deterioration becomes apparent.
4. Extensive validation on a large-scale dataset of 13,478 images, supported by rigorous comparative evaluation against existing state-of-the-art approaches.
5. Design choices that support industrial deployment, emphasizing scalability, interpretability, and computational efficiency to ensure practicality in cold storage, distribution, and retail environments.

3. Proposed Techniques and Algorithm

The system being proposed employs a multimodal learning strategy that combines visual information from RGB images and chemical/environmental information from numerical sensor measurements for spoilage classification of bananas. Since these modalities observe fundamentally different mechanisms of the ripening process; images observe macroscopic visual cues such as browning, color changes, and changes in texture, while numerical sensors observe internal variations in biochemical, gaseous, and environmental parameters, a processing pipeline is created separately for each modality. The processing pipelines are each optimized independently using the most appropriate model families, and merged afterward using a structured confidence-weighted late fusion algorithm.

3.1 Numerical Data Modelling Pipeline

The numerical data contains 27 heterogeneous sensor features: temperature, pressure, humidity, gas sensing (TGS20, TGS02, SGP), and 18 hyperspectral reflectance bands (410 nm to 940 nm). To systematically assess how each feature group contributes to accuracy, the data is analyzed in three configurations:

(1) all features (27),
(2) gas & environmental features (9), and
(3) spectral features (18).

Before model development, numerical labels which were represented in five classes (0–4) were remapped to four classes (0–3) to directly correspond with the image data set, thus eliminating a significant inconsistency in the fusion. Preprocessing was performed using standard methods including cleaning, handling missing values, and standardization via StandardScaler (fitted only on the training split to avoid data leakage). Hyperparameter tuning and model selection were conducted with cross-validation on the training, since no validation set was assigned.

An exhaustive benchmarking procedure was performed for eight machine-learning algorithms: SVM (RBF), Random Forest, 2-layer ANN, 3-layer ANN, KNN, Gradient Boosting, XGBoost, and a 1D-CNN built for tabular data. These methods were selected to present a variety of learning paradigms including distance-based, margin-based (SVM), ensemble-based (RF, GB, XGBoost), deep-learning based (ANN, CNN) and probabilistic tree-based assessment.

Random Forest performed better than every model over the three feature groups. The Random Forest produces the highest accuracy, the most balanced precision-recall scores, and the most stable results across folds of data that were used to train the models. Bootstrapping aggregation, randomized features per split, and resilience to overfitting help Random Forest tell every feature's non-linear interaction within a feature's heterogeneous types. Random Forest even handles the sample size limitation in competing with deep neural networks, which consistently overfit their training dataset despite stopping rules and regularization. Therefore, due to the superior performance of Random Forest compared to other models, Random Forest was selected as the final numerical classifier.

When inference is being done, Random Forest creates a probability distribution

$$P_n = [p_1^n, p_2^n, p_3^n, p_4^n],$$

computed as the normalized frequency of class predictions across all T trees

$$p_c^n = \frac{1}{T}\sum_{i=1}^{T} \mathbf{1}(h_i(x) = c).$$

The confidence score for each modality nnn is defined as:

$$C_n = \max(P_n)$$

where Pn represents the probability distribution predicted by the model for the four classes. The value Cn is later used during the multimodal fusion stage to compute adaptive weights.

3.2. Image Modelling Pipeline

The image dataset represents visual characteristics at the surface level associated with banana spoilage. The images are resized to the input resolution required for EfficientNet architecture and normalized to reduce variability during training. The model was trained using augmentation (rotation, controlled brightness variation, horizontal flips, and an added mild geometric distortions) only on the training split, which helps further reduce overfitting and improve generalization due to a relatively small dataset.

The modelling framework started with baseline experiments such as PCA + Logistic Regression and custom CNN architectures, with a custom CNN yielding respectable accuracy. However, the greatest accuracy was achieved using EfficientNet-B0, a state-of-the-art convolutional neural network that maintains balance across network depth, width, and resolution collectively using the technique of compound scaling. With pretrained ImageNet weights, the model was trained iteratively assigned to optimize learning in two phases: an initial short warm-up training with image data in which the convolutional layers were frozen first followed by the later part of model training consisting of thawed convolutional layers with the ability to fine-tune the top section of the backbone. The model demonstrated strong convergence behaviour and showed effective predictive class separation characteristics.

EfficientNet learns hierarchical representations that depict attribute change due to varied stages of spoilage at the end of these representations depicting degrading color, bruising, localized dark patches and overall surface texture. This is a desirable aspect in identifying ripeness or spoilage characteristics throughout the ripening period. The model output is in the form of a softmax probability vector.

$$P_v = [p_1^v, p_2^v, p_3^v, p_4^v],$$

The confidence score for the visual stream is computed as

$$C_v = \max(P_v),$$

Where Pv denotes the class-probability distribution predicted by the visual model. Based on its superior accuracy, calibration behaviour, and overall stability across experiments, EfficientNet-B0 was selected as the final image classifier for the visual modality.

3.3. Fusion Technique: Confidence-Weighted Late Fusion

Once both pipelines have offered their output, the system utilizes late fusion to leverage the strength of both numerical and visual data. Late fusion is more advantageous than early fusion, since both modalities have vastly different underlying structure, scale, and sampling characteristics where the reliability of training together at the feature level is not possible.

To ensure that both modalities contribute proportionally to their confidence levels, a **confidence-weighted fusion** method is implemented. The weights for each modality are calculated as:

$$w_v = \frac{C_v}{C_v + C_n}, \qquad w_n = \frac{C_n}{C_v + C_n}.$$

The final probability distribution is then computed as a convex combination of the two outputs:

$$P_{\text{final}} = w_v \cdot P_v + w_n \cdot P_n.$$

The predicted class label is:

$$\hat{y} = \arg\max P_{\text{final}}.$$

3.3.1 Dynamic Adaptation Behaviour of the Fusion Strategy

The confidence-weighted fusion model allows the system to fluctuate its reliance between the visual and numerical models in a way that reflects how one would naturally rely on different forms of input evidence. When the image input is clear, well lit, and shows the banana's surface in perfect detail, for instance, the visual model is very confident in its prediction. Its probability outputs favor one class strongly, meaning a color transition, texture change, or visible bruising gives reliable evidence that the current banana is a banana. In this state, the fusion process also gives more weight to the visual stream, allowing the final chosen class to rely on the strength of the image information.

Further, the balance of reliance may shift as the quality of the image decreases. Other factors such as shadows and glare, motion blur, or just an odd viewing angle make visual characteristics more difficult for humans to interpret and thus decrease the confidence position of the EfficientNet model. But, at the same time, the numerical model often provides stable results, as we see the chemical sensors and environmental measures are often not fooled by visual artifacts, even if the visual evidence is low-quality. Thus, in a down-refusers situation, the fusion model increases the influence of the numerical predictions while decision-making remains valid.

Conversely, and perhaps most robustly, if both models return a confident output for the same class independently, then the agreement is especially compelling, as those predictions are likely to be stable and robust in agnostic decision contexts.

3.4. Complete Algorithmic Workflow:

The system executes the following steps:

1. Input Acquisition
Capture RGB image.
Collect sensor readings (27 numerical features).

2. Numerical Preprocessing
Clean, scale, and remap labels.
Apply trained Random Forest to compute PnP_nPn and CnC_nCn.

3. Image Preprocessing
Resize, normalize, and (during training) augment.
Apply EfficientNet-B0 to compute PvP_vPv and CvC_vCv.

4. Fusion
Compute modality weights using confidence levels.
Generate fused probability distribution.
Produce final class prediction.

5. **Output**
Display final class, fused confidence score, and both modality probabilities.

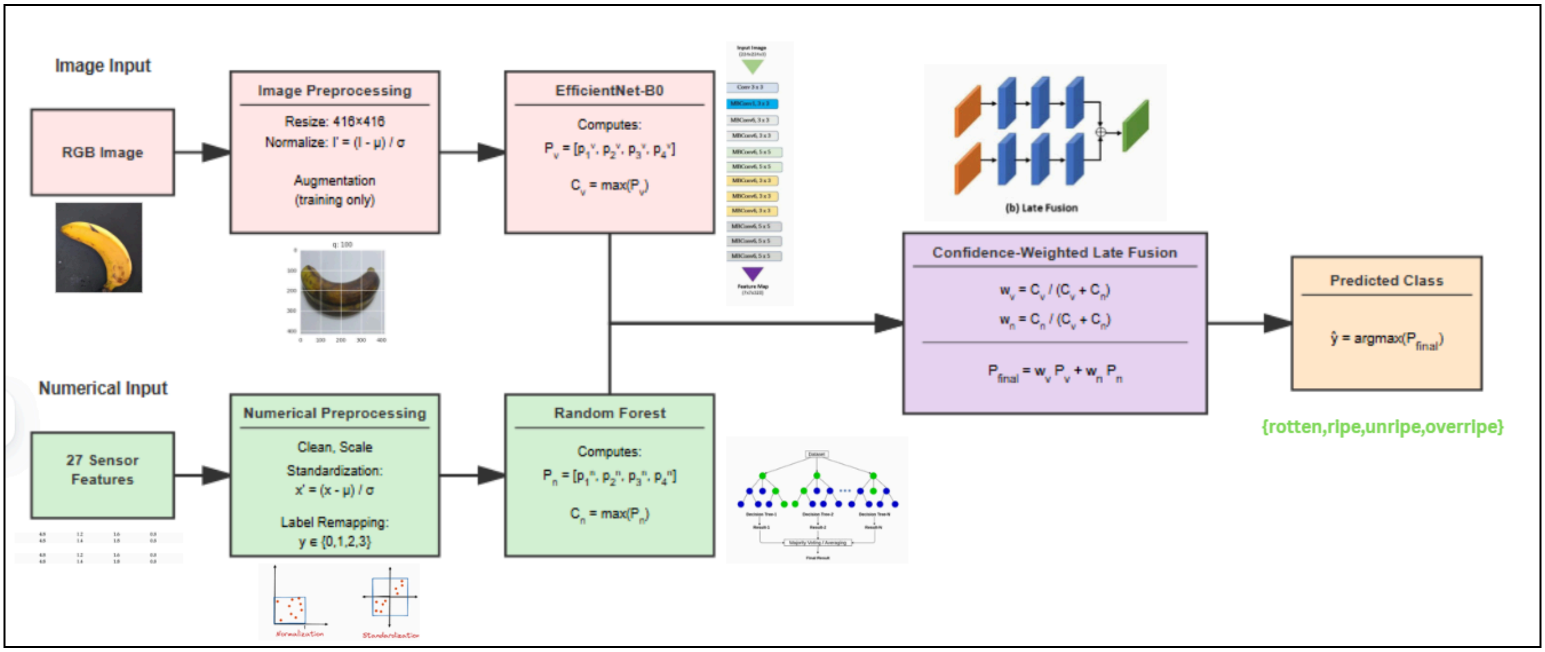


Fig. The Proposed Architecture

4. Datasets Used

This work uses a dual-modality dataset that contains (i) a large-scale banana image dataset collected from Kaggle and (ii) a numerical sensor dataset collected from a previous study. These datasets provide an effective multimodal classification system leveraging visual degradation signals with biochemical and environmental information.

4.1 Image Dataset - Banana Ripeness Images :

The visual component of this study is based on the Banana Ripeness Classification Dataset obtained from Kaggle, containing 13,478 high-resolution RGB images representing four distinct ripeness categories:

1. Unripe
2. Ripe
3. Overripe
4. Rotten

Image Collection Characteristics

- Images captured under diverse real-world conditions including varying lighting, angle, and background variations.
- Each image shows a single banana specimen, enabling both pixel-level and shape-based analysis.
- Resolution varies, but all images were standardized to 416×416 px through preprocessing.

Dataset Split

The dataset was divided into stratified subsets according to standard machine learning practice:

Subset	Samples	Purpose
Training	11,793 (87%)	Feature learning & model training with augmentation
Validation	1,123 (8%)	Hyperparameter tuning & early stopping
Test	562 (4%)	Final evaluation & multimodal fusion

Class Distribution

1. **The distribution of classes still reflects the natural ripening progressions:**

   Rotten - ~4,500 (33.4%)
   Ripe - ~4,000 (29.7%)|
   Overripe - ~2,600 (19.3%)
   Unripe - ~2,200 (16.3%)

2. **Preprocessing and Augmentation**

   Auto-orientation with the EXIF
   dataResize to 416×416
   Normalize pixel values to [0,1]
   Augmentation only for training: flips, rotations, zoom, changes in brightness/hue, and blur

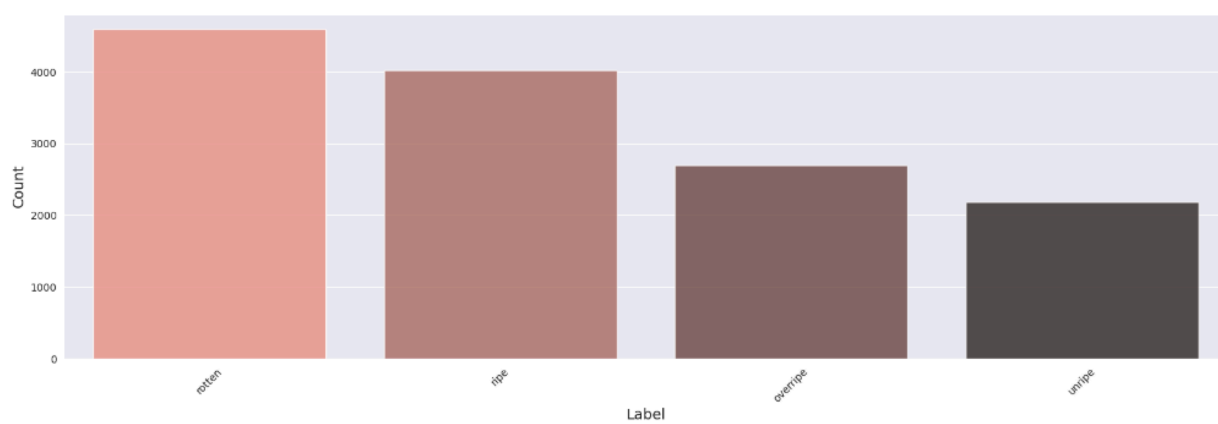


Fig. The Banana Ripeness dataset Visualization

4.2 Numerical Sensor Dataset:

The numerical dataset, adopted from a peer-reviewed research publication, contains 27 sensor features per sample, describing chemical, environmental, and spectral properties of banana specimens. This dataset is completely independent of the Kaggle image dataset but provides complementary biochemical information aiding multimodal fusion.

Raw Attributes (27 Features)

The features include:

- **Environmental (6 features)**
  Temp-int, Temp-ext, Humid-int, Humid-ext, Press-int, Press-ext
- **VOC Sensors (3 features)**
  TGS20, TGS02, SGP
- **Hyperspectral (18 features)**
  SpA410 → SpL940 (18 spectral bands covering 410–940 nm)

Dataset Split

The dataset is already split into the following:

1.Training Set

2. Testing Set

There was no validation set explicitly provided. however, cross-validation (k = 5) was performed on the training set for hyperparameter tuning.

Label Harmonization

The original dataset had 5 classes (0-4), while the image dataset had 4 classes (0-3). To facilitate fusion:→ All numerical labels were scored into 0-3 in order to avoid inconsistency within multimodal prediction.

Preprocessing Steps

1. StandardScaler fitted on training set alone.
2. Applied same scaling to test set (as to not get leakage of data).
3. Dimensionality reduction via PCA (18 spectral features → 6 PCs).
4. Removal of outliers (Isolation Forest).
5. Median imputation for rare missing values.

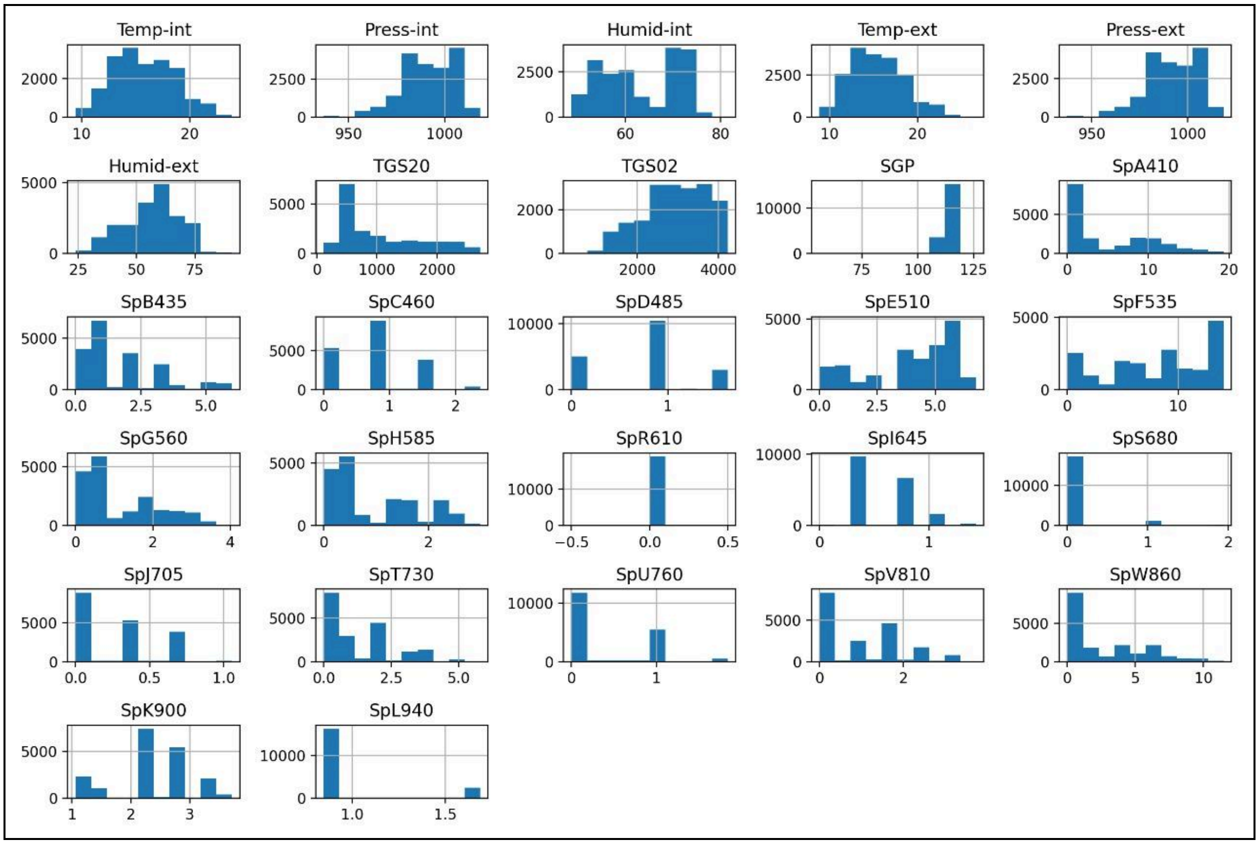


Fig: Numerical Dataset and values Overview

5. Experiments and Results

This section details the experimental procedures carried out as part of assessing individual modality-specific models and the final multimodal fusion framework. Experiments were completed using the image dataset separately from the numerical dataset, followed by late-fusion evaluation. Each model was trained, validated, and evaluated using the partitions from the datasets mentioned earlier to ensure methodological consistency and reproducibility.

5.1 Image-Based Experiments

Three progressively advanced models were implemented to benchmark visual classification performance. Each model was trained using the same training/validation split and evaluated on the 562-image test set to allow direct comparison.

5.1.1 Baseline Image Classifier: Logistic Regression

The first experiment established a classical machine-learning baseline using **multinomial logistic regression**. To enable compatibility with linear models, images were preprocessed as follows:

- Input images resized to 416×416 pixels
- Converted into grayscale
- Flattened into a 1-D feature vector
- Principal Component Analysis (PCA) applied to reduce dimensionality (retaining 95% variance)

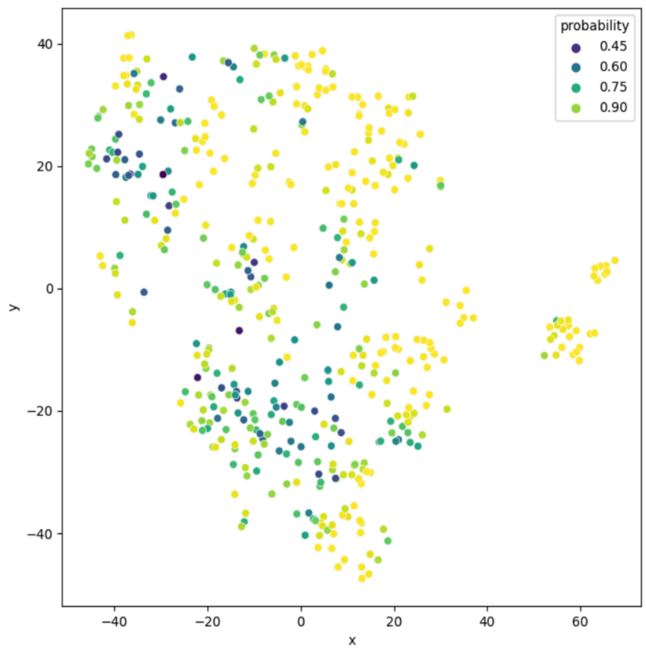
The logistic regression classifier used an L2-regularized softmax objective. This baseline quantifies the complexity of the visual decision boundary by demonstrating how well a linear model can separate ripeness classes.

RESULTS:

```
model fit in 230 iterations
accuracy: 0.8917
f1: 0.8920
              precision    recall  f1-score   support

    overripe       0.91      0.88      0.89       120
        ripe       0.81      0.84      0.83       120
      rotten       0.88      0.89      0.89       120
      unripe       0.97      0.96      0.96       120

    accuracy                           0.89       480
   macro avg       0.89      0.89      0.89       480
weighted avg       0.89      0.89      0.89       480
```



3.1.2 Convolutional Neural Network (CNN)

The second experiment introduced a custom **deep convolutional neural network**, designed to capture spatial patterns and texture cues associated with ripeness.

Architecture highlights:

- Four convolutional blocks (Conv → BatchNorm → ReLU → MaxPool)
- Two fully connected layers
- Softmax classification head

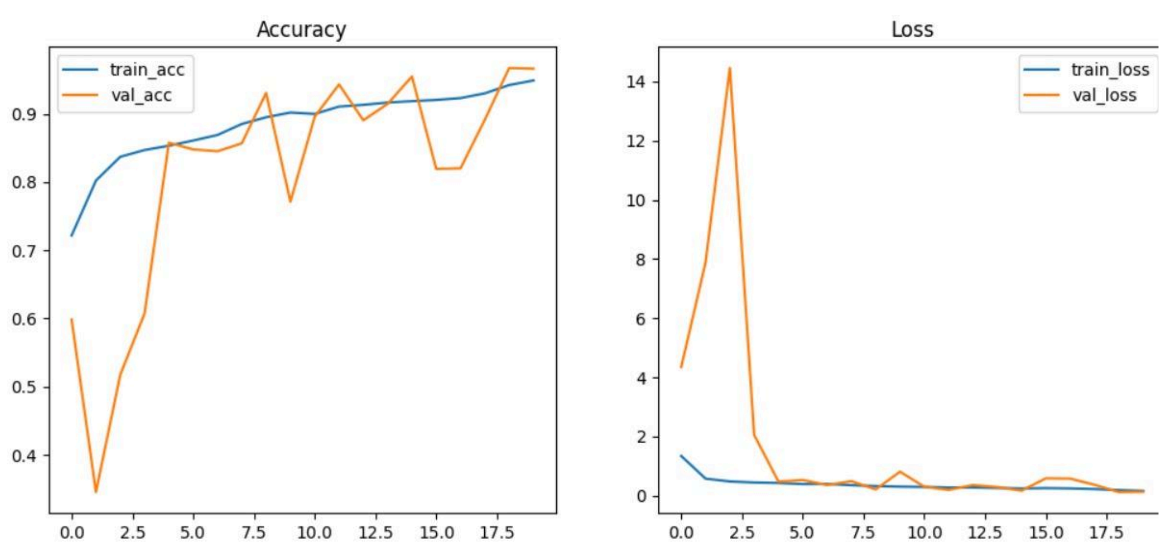
Training employed:

- Adam optimizer
- Learning-rate scheduling
- Cross-entropy loss
- Early stopping to prevent overfitting

Layer (type)	Output Shape	Param #
conv2d (Conv2D)	(None, 126, 126, 32)	896
batch_normalization (BatchNormalization)	(None, 126, 126, 32)	128
max_pooling2d (MaxPooling2D)	(None, 63, 63, 32)	0
conv2d_1 (Conv2D)	(None, 61, 61, 64)	18,496
batch_normalization_1 (BatchNormalization)	(None, 61, 61, 64)	256
max_pooling2d_1 (MaxPooling2D)	(None, 30, 30, 64)	0
conv2d_2 (Conv2D)	(None, 28, 28, 128)	73,856
batch_normalization_2 (BatchNormalization)	(None, 28, 28, 128)	512
max_pooling2d_2 (MaxPooling2D)	(None, 14, 14, 128)	0
flatten (Flatten)	(None, 25088)	0
dense (Dense)	(None, 256)	6,422,784
dropout (Dropout)	(None, 256)	0
dense_1 (Dense)	(None, 4)	1,028

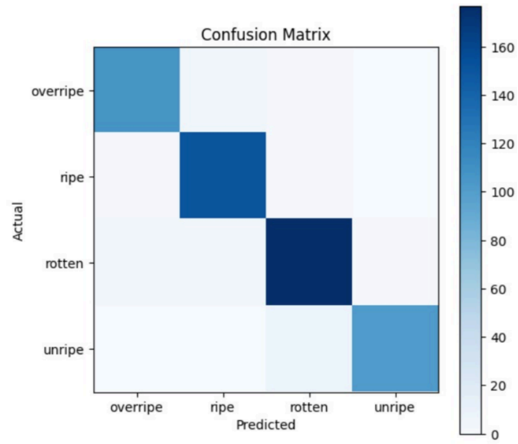
This experiment evaluated the benefit of introducing hierarchical feature extraction over the simpler linear baseline.

RESULTS:



Classification Report:

	precision	recall	f1-score	support
overripe	0.95	0.95	0.95	113
ripe	0.95	0.98	0.96	154
rotten	0.95	0.96	0.95	185
unripe	0.99	0.93	0.96	110
accuracy			0.96	562
macro avg	0.96	0.95	0.96	562
weighted avg	0.96	0.96	0.96	562



5.1.3 EfficientNet-B0 Fine-Tuned Model

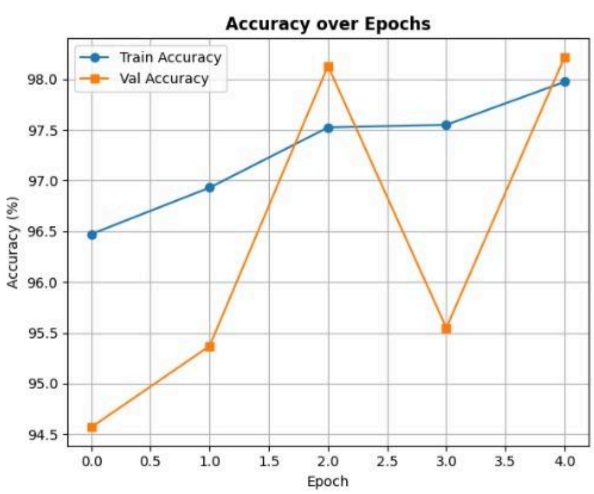
The final image experiment used EfficientNet-B0, a state-of-the-art convolutional architecture pre-trained on ImageNet. Fine-tuning was conducted in two phases:

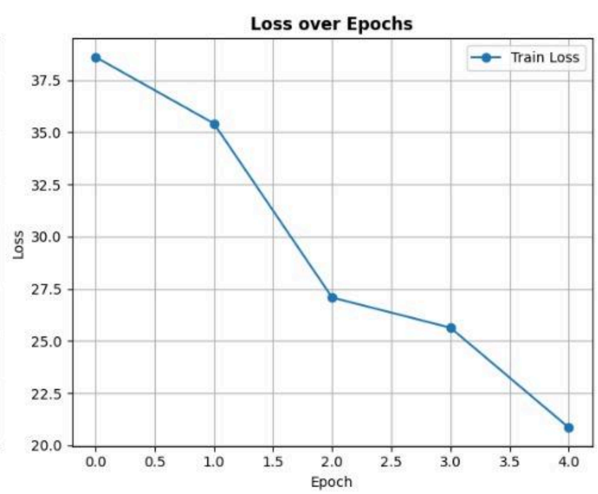
1. Phase 1 – Head Training:
   Base layers frozen & classification head trained on banana images.

2. Phase 2 – Partial Unfreezing:
   Top 40% of EfficientNet layers unfrozen; full model fine-tuned with a reduced learning rate.

This experiment explored transfer learning benefits, leveraging pretrained visual representations to improve ripeness classification accuracy and generalization.

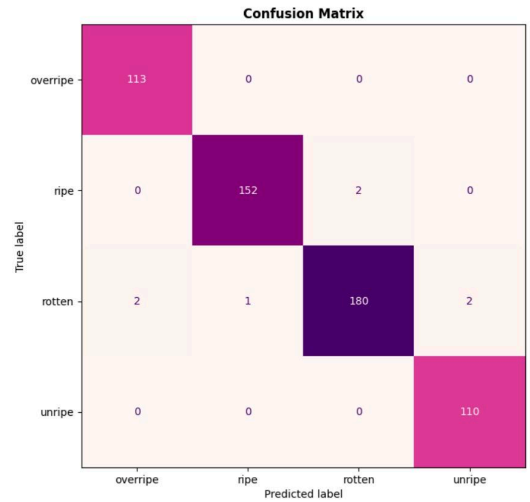
RESULTS:





Model Evaluation Metrics: Classification Report

	precision	recall	f1-score	support
overripe	0.97	0.97	0.97	113
ripe	0.96	0.99	0.97	154
rotten	0.99	0.95	0.97	185
unripe	0.98	0.99	0.99	110
accuracy	0.98	0.98	0.98	1
macro avg	0.98	0.98	0.98	562
weighted avg	0.98	0.98	0.98	562



This model gave the best accuracy, hence it was chosen as the image side model for our architecture

5.2 Experiments with Numerical Data

Eight distinct machine-learned models were trained against the 12-dimensional numerical data set, which consisted of hyperspectral PCs, VOC sensor data, and environmental features. Each model took standardized inputs and was assessed on the same cross-validation folds for the sake of equitable and reproducible benchmarking.

Each model was trained and tested with three different feature splits:
• All Features (the complete 12-dimensional representation)
• Gas & Environmental Features (6 features from VOC and environmental sensors)
• Spectral Features
The numerical experiment suite consisted of:

5.2.1 Support Vector Machine (RBF Kernel) : The SVM model was included due to its capacity to model non-linear class boundaries using kernel transformations over low-dimensional feature representations.

5.2.2 Random Forest Classifier: A 500-tree ensemble model was evaluated in its viability at overcoming noise, interaction between features, multicollinearity, and small sample size respectively - making it a well-suited model for the structured sensor data.

5.2.3 Artificial Neural Network (2-Layer MLP): A shallow neural network architecture was used to understand if moderate non-linear thresholding would improve test set performance on tabular data generation.

5.2.4 Artificial Neural Network (3-Layer MLP): A deeper variation was created to provide context on whether added depth could positively impact learning representations from biochemical and environmental features.

5.2.5 K-Nearest Neighbour: A lazy learning algorithm a model was tested on to understand the raw geometric class separation of classes under standard feature space.

5.2.6 Gradient Boosting Machine: A boosted tree model capable of capturing complex interactions and sequentially correcting weak learners.

5.2.7 XGBoost Classifier: A highly optimized gradient-boosting algorithm tested for its ability to generalize well on relatively small datasets.

5.2.8 1D-CNN for Tabular Data: A convolution-based model in which numerical features are restructured into 1-D sequences, enabling the capture of local dependency structures otherwise missed by traditional models. These eight experiments provided a comprehensive evaluation of classical, ensemble-based, neural, and convolutional methods. Random Forest ultimately emerged as the most stable, accurate, and balanced classifier across all three feature subsets.

RESULT ANALYSIS :

Delimiter: ,

	Model	Accuracy	Precision	Recall	F1 Score
1	random_forest_model_s.pkl	0.9997520456236053	0.9997521224372287	0.9997520456236053	0.9997520456116925
2	xgboost_model_s.pkl	0.9992561368708158	0.9992572864613384	0.9992561368708158	0.9992562131844711
3	gradient_boost_model_s.pkl	0.9987602281180263	0.9987631407682689	0.9987602281180263	0.9987598011344069
4	knn_model_s.pkl	0.9884701214976445	0.9885650332768283	0.9884701214976445	0.988474112012715
5	cnn_model_s.h5	0.9778080833126704	0.9779206586773308	0.9778080833126704	0.9777857631023793
6	ann_3layer_model_s.keras	0.9771881973716836	0.9777503952995117	0.9771881973716836	0.9772348744558761
7	ann_2layer_model_s.keras	0.9717332010909993	0.9721087362124589	0.9717332010909993	0.9717286732563766
8	svm_rbf_model_s.pkl	0.8239523927597322	0.828509145269831	0.8239523927597322	0.8252759883157186

Model comparison of ‘all’ feature models

	Model	Accuracy	Precision	Recall	F1 Score
1	random_forest_model_s.pkl	0.999008182494421	0.9990103273452386	0.999008182494421	0.9990077950767824
2	xgboost_model_s.pkl	0.999008182494421	0.9990084120301844	0.999008182494421	0.9990081053472913
3	gradient_boost_model_s.pkl	0.9987602281180263	0.9987632936834294	0.9987602281180263	0.9987598394342581
4	knn_model_s.pkl	0.9873543268038681	0.9874810449888025	0.9873543268038681	0.9873585348043183
5	ann_3layer_model_s.keras	0.9699975204562361	0.9705831033013982	0.9699975204562361	0.9700138622292631
6	cnn_model_s.h5	0.9485494668980907	0.9514356988558799	0.9485494668980907	0.9484653424316349
7	ann_2layer_model_s.keras	0.9453260600049591	0.9484894220276391	0.9453260600049591	0.9450101460532658
8	svm_rbf_model_s.pkl	0.8193652367964295	0.8245007518599835	0.8193652367964295	0.8208604104439511

Model comparison of ‘gas&env’ feature models

	Model	Accuracy	Precision	Recall	F1 Score
1	random_forest_model_s.pkl	0.9830151252169601	0.9830874310814778	0.9830151252169601	0.9830166275995874
2	xgboost_model_s.pkl	0.9830151252169601	0.9831044223074681	0.9830151252169601	0.9830335090604908
3	gradient_boost_model_s.pkl	0.9784279692536574	0.9784995295662967	0.9784279692536574	0.9784397047911854
4	cnn_model_s.h5	0.9747086536077362	0.9750433842089397	0.9747086536077362	0.974716162162071
5	knn_model_s.pkl	0.9737168361021572	0.9740139582293083	0.9737168361021572	0.9737692458155675
6	ann_2layer_model_s.keras	0.9695016117034466	0.9701043056394834	0.9695016117034466	0.9696374165317332
7	ann_3layer_model_s.keras	0.9675179766922887	0.9679941378658479	0.9675179766922887	0.9675707783529945
8	svm_rbf_model_s.pkl	0.9474336722043144	0.9489880626953007	0.9474336722043144	0.9473642458913223

Model comparison of 'specter' feature models

Random Forest was chosen as the top numeric classifier for all three categories because:

1. It achieved the highest accuracy among all models.
2. It maintained balanced precision, recall, and F1-score across all classes.
3. It handled noise, feature interactions, and small sample size better than other methods.
4. It remained the most stable model across cross-validation folds.

These characteristics made it the optimal numeric component for the final multi-modal fusion.
More thorough analyses of the top models for each of the following categories:

```
=== Random Forest - Test Summary ===
Accuracy  : 0.9998
Precision : 0.9998
Recall    : 0.9998
F1 Score  : 0.9998
```

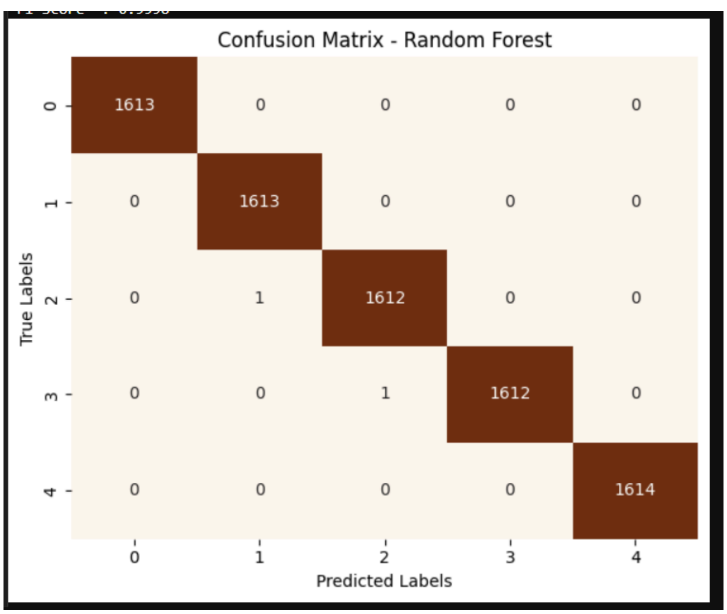


Fig: Evaluation metrics of the best 'all' feature model (random forest)

```
=== Random Forest - Test Summary ===
Accuracy  : 0.9990
Precision : 0.9990
Recall    : 0.9990
F1 Score  : 0.9990
```

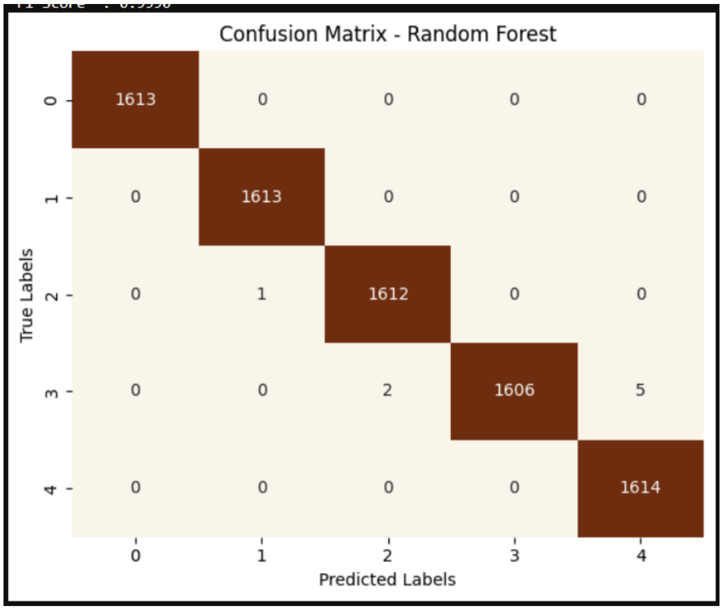


Fig: Evaluation metrics of the best ‘gas&env’ feature model (random forest)

```
=== Random Forest - Test Summary ===
Accuracy  : 0.9830
Precision : 0.9831
Recall    : 0.9830
F1 Score  : 0.9830
```

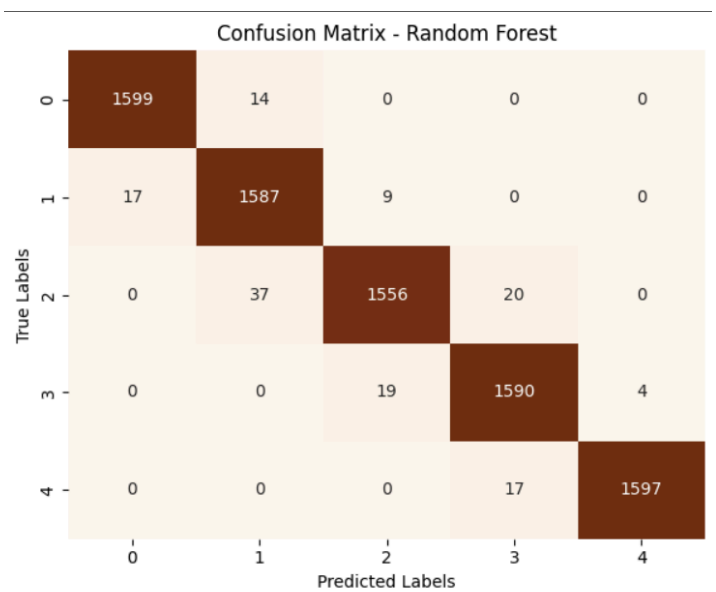


Fig: Evaluation metrics of the best ‘specter’ feature model (random forest)

Overall Comparison:

Model	Dataset	Accuracy	Precision	Recall	F1 Score
gass&env_random_forest_model	numerical	0.999	0.999	0.999	0.999
gass&env_xgboost_model	numerical	0.999	0.999	0.999	0.999
gass&env_gradient_boost_model	numerical	0.999	0.999	0.999	0.999
gass&env_knn_model	numerical	0.987	0.987	0.987	0.987
gass&env_ann_3layer_model	numerical	0.97	0.971	0.97	0.97
gass&env_cnn_model	numerical	0.949	0.951	0.949	0.948
gass&env_ann_2layer_model	numerical	0.945	0.948	0.945	0.945
gass&env_svm_rbf_model	numerical	0.819	0.825	0.819	0.821
specter_random_forest_model	numerical	0.983	0.983	0.983	0.983
specter_xgboost_model	numerical	0.983	0.983	0.983	0.983
specter_gradient_boost_model	numerical	0.978	0.978	0.978	0.978
specter_cnn_model	numerical	0.975	0.975	0.975	0.975
specter_knn_model	numerical	0.974	0.974	0.974	0.974
specter_ann_2layer_model	numerical	0.97	0.97	0.97	0.97
specter_ann_3layer_model	numerical	0.968	0.968	0.968	0.968
specter_svm_rbf_model	numerical	0.947	0.949	0.947	0.947
all_random_forest_model	numerical	0.999	0.999	0.999	0.999
all_xgboost_model	numerical	0.999	0.999	0.999	0.999
all_gradient_boost_model	numerical	0.999	0.999	0.999	0.999
all_knn_model	numerical	0.988	0.989	0.988	0.988
all_cnn_model	numerical	0.978	0.978	0.978	0.978
all_ann_3layer_model	numerical	0.977	0.978	0.977	0.977
all_ann_2layer_model	numerical	0.972	0.972	0.972	0.972
all_svm_rbf_model	numerical	0.824	0.829	0.824	0.825
Logistic Regression model	Image	0.89208	0.89	0.89	0.89
Convolutional Neural Network	image	0.96	0.96	0.96	0.96
Effecient Net model	image	0.979	0.979	0.97	0.97

Fig: Evaluation metrics of all the experiments performed

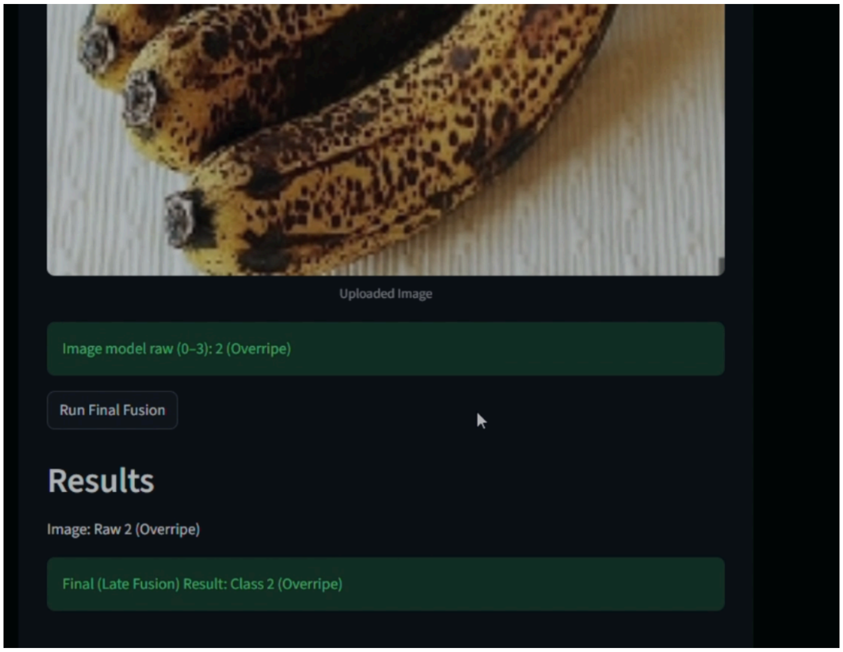
5.3 Multimodal Fusion Experiment

After identifying the best performing model in each modality - EfficientNet-B0 for imagery, and Random Forest for numerical - a multimodal experiment was conducted to assess the performance of the fused model. Both models produced independent class probability vectors and confidence levels were used to create normalized weights for each modality. These weights were then used to calculate a fused probability distribution.

This experiment evaluated:

1. Whether fusion increases classification dependability beyond either modality alone.
2. How confidence weighting varies with image quality and sensor consistency.
3. Whether fused predictions exhibit less variance and more agreement across ripeness classes.

This multimodal experiment supports the proposed solution by demonstrating how combined visual and sensor accuracy improves ripeness identification even better than either modality alone.




```
Your Streamlit app is live at: NgrokTunnel: "https://eunice-feastless-biserially.ngrok-free.dev" -> "http://localhost:8501"
```

5.4 Error Analysis

1. Class Inconsistency Between Modalities

There were five classes in the numerical dataset and four in the image dataset.
This mismatch caused a classification alignment error (Error 53.1) during fusion.
By remapping numerical labels into a single 0–3 class range, this was fixed, guaranteeing

2. Early Stopping and Overfitting

Due to the small dataset size, ANN and 1D-CNN models were vulnerable to overfitting. Early stopping decreased overfitting, but it occasionally caused training to end too soon.

3. Model-Specific Weaknesses

- KNN struggled with high-dimensional data and was highly sensitive to feature scaling.
- ANN (3-layer) and 1D-CNN required more training samples to converge properly.
- Gradient Boosting tended to overfit minority classes.
- SVM (RBF) was computationally heavy and less stable across folds.
- XGBoost showed good performance but lacked the consistency of Random Forest.

4. Image Model Misclassification

EfficientNet occasionally confused visually similar classes (e.g., mid-stage ripeness). Uneven lighting conditions also created minor inconsistencies despite augmentation.

6. Conclusion

This study shows that combining image data with numerical sensor measurements greatly improves the accuracy of banana spoilage prediction. While EfficientNet captured visual degradation cues, traditional machine-learning models captured biochemical and environmental characteristics. The system produced a robust and dependable multimodal classification when Random Forest, the strongest numerical model, was integrated with the image model via late fusion. The outcomes confirm that multi-modal learning is useful for evaluating agricultural quality.

7. Future Work

This project may be expanded in the future to include:

- Adding more fruit and vegetable varieties to the dataset, particularly those with few publicly available datasets.
- Enhancing the model generalization by incorporating bigger multi-modal datasets.
- Investigating cutting-edge fusion methods like attention-based cross-modal transformers.
- Improving domain adaptation and illumination normalization for stronger visual classification.

7.1 Work Update

Using both numerical and image-based datasets, the project has effectively created a banana spoilage detection system. The work offers a thorough understanding of spoilage progression by combining these two modalities, showing how visual cues and numerical sensor-like features can work well together. The project is positioned as a solid baseline for multi-modal food quality assessment thanks to this dual approach.

7.2 Work Completed

The majority of current research on food spoilage detection focuses on image-based techniques, primarily due to the greater accessibility of visual datasets for fruits and vegetables. **Numerical data, on the other hand, has gotten much less attention**. Examples of this include chemical readings, temperature-based parameters, and sensor outputs. There are very few publicly available numerical datasets for systematic model development, and only a small number of studies have experimented with numerical features.

7.3 Work Remaining

A significant area for further investigation is the numerical aspect of food spoilage research. Building sizable, well-organized numerical datasets, creating reliable multi-modal fusion models, and validating them across a variety of food categories, including milk, fruits, and vegetables, are all very promising. Developing this unexplored field can result in practical, useful solutions that use inexpensive, sensor-driven systems to support supply chain monitoring, quality control, and early spoilage detection.

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